

An Efficient Blockchain-based Cross-domain Authentication and Secure Certificate Revocation Scheme

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Abstract—The existing cross-domain authentication mechanisms are established based on well-known Public Key Infrastructure (PKI) systems, where digital certificates issued by Certificate Authority(CA) serve to authenticate the identity of entities. The complex authentication path, accompanied by multiple signatures and verifications, causes low authentication efficiency. In addition, resulting from an "update gap" between Certificate Revocation List(CRL) and Online Certificate Status Protocol(OCSP) mechanism, the revoked certificates will be exposed to DDoS attacks. For addressing aforementioned two challenges, an efficient blockchain-based authentication and secure certificate revocation scheme is proposed. Based on the non-tamperability and traceability of the blockchain, the signature module of X.509 digital certificate is replaced by the certificate hash value which serves as the trust certificate between domains. Furthermore, a consensus algorithm is adopted for random number hash broadcasting to improve efficiency. Theoretical analysis shows the proposed scheme has the security characteristics of non-repudiation, anonymity and anti DDoS attack. The experimental results demonstrate the proposed scheme has the advantage over existing cross-domain authentication and certificate revocation scheme at communication and computing cost and security respectively.

Keywords—blockchain; consensus algorithm; cross-domain authentication; certificate revocation

I. INTRODUCTION

Digital certificate[1], a digital file issued by a trusted CA, is composed of public key owner's information, public key, expiration date, and extension information, which is utilized to identify individuals, devices or other entities involved in communication on the Internet. Through the digital certificate, internet users' identity can be verified. During the process, the digital certificate provided by the user can be decrypted according to the CA public key and a successful decryption indicates the validity of the certificate and the reliability of the user. Digital certificates are widely applied in e-commerce, mobile payment, secure e-mail and other scenarios.

In the distributed network environment, organizations set up authentication servers to form a relatively independent trust domain in order to prevent unauthorized users from accessing internal shared resources, as shown in Figure 1. However, Internet users require multiple network services

which cannot be requested in a single independent trust domain, so cross-domain authentication problems arise[2]. When performing cross-domain authentication, a user commonly obtain other users' certificate through a complex certification path, which constitutes multiple certificate transmissions, digital signature and decryption. Correspondingly, frequent cross-domain access will lead to increased network delay and decreased throughput.

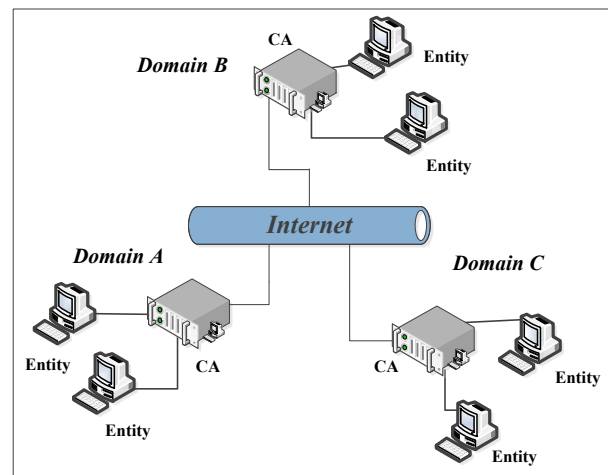


Figure 1. Coexistence model of multiple independent trust domains.

Each certificate has a validity period. Usually, the new certificate will be issued and updated before the old certificate expires. However, when the user's private key or the CA certificate is considered insecure, or the user no longer trusts their CA, the certificate have to be revoked in advance[3]. The X.509 standard is a widely accepted scheme to standardize the format of digital certificate in most network security applications. The standard stipulates that each CA must issue regularly a CRL to record those unexpired certificate that has been revoked by CA. When the user receives a digital certificate in a message, he must query and verify whether the certificate has been revoked in CRL. The CRL mechanism is vulnerable to man-in-middle attacks and DoS attacks. The periodicity of CRL update will inevitably lead to the situation that revoked certificates are not updated to CRL in time [4], so that attackers can make use of the expired digital certificate to request services or access resources. Moreover, when communication exists

between multiple domains each of which maintains a separate CRL, these CRLs must be shared in time for security authentication. Therefore, it is an important direction to realize efficient cross-domain authentication and share certificate revocation lists in different domains.

Zhang et al.[5] proposed a generalized virtual enterprise cross-domain authentication scheme based on virtual bridge CA using a non-trusted center elliptic curve threshold signature and a variable multi-party protocol. However, threshold signature causes high interaction cost by splitting key factor, and the scalability of user joining and canceling is low. Jie Z et al.[6] proposed a multi domain joint authentication protocol, which uses blind signature to realize mutual authentication between entities in different domains and ensures the security of sharing resources among multiple domains. However, during cross-domain access, the time consumed by blind signature algorithm increases the delay of access resources and reduces the communication efficiency. In addition, many certificate revocation mechanisms have been proposed. Based on the game theory model, a new voting based security scheme is designed[7], which can respond quickly according to the current system conditions. However, there are obvious multi-party computing security problems that have not been resolved in this scheme. Chariton et al.[8] proposed a new low latency solution to solve the problem of high cost of CRL and some browsers give up checking the revocation status of certificates based on the existing scalable and high-performance infrastructure of DNS, it can quickly provide the latest and accurate revocation information of certificates.

Blockchain technology[9] has been widely applied in distributed networks since Nakamoto proposed the concept of bitcoin. Blockchain technology plays a great role in promoting the development and application of digital certificates. Several blockchain-based digital certificate authentication and revocation schemes have been proposed one after another. A decentralized PKI authentication system based on bitcoin blockchain system was proposed[10], where Certcoin instead of CA offered efficient key query and identity retention functions. However, there existed the problem of user privacy disclosure caused by using blockchain public general ledger to directly record user identity and public key binding. Based on the alliance chain technology, the root CA served as the node in the blockchain network, and a cross-domain authentication protocol was designed[11]. However, due to the limitations of the consensus algorithm of the blockchain, the authentication efficiency was reduced. Besides, the scheme did not provide a certificate revocation mechanism. Zhang et al.[12] utilized the characteristics of the decentralized consensus mechanism and introduced a key sharing scheme to achieve the collaborative management of CRLs by multiple CAs.

At present, these studies have not given a complete solution that can simultaneously solve cross-domain authentication and certificate revocation. In view of the above problems and the improvement of existing research results, this paper proposes a digital certificate scheme based on blockchain to achieve efficient cross-domain authentication and certificate security revocation. The rest of

the article is arranged as follows: Section II mainly introduces the basic knowledge of authentication and the improvement of digital certificate content. Section III gives the system architecture of the solution, the consensus algorithm, and the specific cross-domain authentication protocol and certificate security revocation protocol. Section IV analyzes the security and feasibility of the proposed scheme. Finally, the whole paper is summarized and future work directions are presented.

II. DIGITAL CERTIFICATE

A. Design of Certificate

The X.509 standard is a widely accepted digital certificate scheme. The public key certificate related to each user is crucial to X.509, which is created by CA and saved to directory server by CA or user.

The digital certificate designed in the proposed scheme is based on the X.509 standard, where the signature algorithm and signature module mentioned above are eliminated. Because the blockchain has the characteristics of non-tamperability, traceability, and common maintenance by multiple parties, the root CA of each domain can be viewed as the node in a blockchain network, and the hash value of the generated certificate is recorded in the blockchain Ledger as the multi domain trust certificate, which can replace the complex signature process. The authentication server of each domain can replace the signature verification process by checking the multi domain trust certificate as the digital certificate verification process.

B. Authentication

The traditional CA certificate operation process is that after the applicant applies for the certificate, the CA verifies the information and issues the digital certificate to the applicant. The specific process is as follows:

Procedure of certificate application

1. The user generates his own key pair and sends the public key and personal information to the CA to apply for a digital certificate.
2. After the CA confirms the identity of the applicant, it signs the applicant's personal information, public key, etc. and sends it to the applicant as a digital certificate:

$$CA = E(PR_{auth}, [T || ID_A || PU_A]) \quad (1)$$
 PR_{auth} is the CA's private key, PU_A is the applicant's public key, ID_A is the applicant's ID, T is other information.
3. After receiving the certificate, the applicant can conduct relevant communication activities.

When communicating in the same trust domain, user A passes the digital certificate he holds to any other participant B in the domain. B uses the CA's public key to verify and read the digital certificate:

$$\begin{aligned} D(PU_{auth}, CA) &= D(PU_{auth}, E(PR_{auth}, [T || ID_A || PU_A])) \\ &= [T || ID_A || PU_A] \end{aligned} \quad (2)$$

The digital certificate is signed with the CA's private key. Therefore, when user B uses the CA's public key to read the certificate content, it also verifies that the digital certificate is indeed from the CA and is reliable.

When communicating across domains, a complex authentication process usually occurs. Suppose A has the digital certificate issued by CA named X1, and B has the digital certificate issued by CA named X2, then A must obtain the public key of X2 to verify the digital certificate of B, which requires that two CAs can exchange their public keys safely. The specific process is as follows.

Procedure of cross-domain authentication

1. X1 issues a digital certificate to X2 signed with X1's private key:

$$CA_{X1,X2} = E(PR_{X1}, [T||ID_{X2}||PU_{X2}]) \quad (3)$$

PR_{X1} is X1's private key, ID_{X2} is X2's ID, PU_{X2} is X2's public key.

2. A uses the public key of X1 to verify and read the digital certificate issued by X1 to X2 to obtain the public key of X2:

$$\begin{aligned} & D(PU_{X1}, CA_{X1,X2}) \\ &= D(PU_{X1}, E(PR_{X1}, [T||ID_{X2}||PU_{X2}])) \\ &= [T||ID_{X2}||PU_{X2}] \end{aligned} \quad (4)$$

PU_{X1} is X1's public key.

3. A uses the obtained X2's public key to verify and read B's digital certificate:

$$\begin{aligned} & D(PU_{X2}, CA_{X2,B}) \\ &= D(PU_{X2}, E(PR_{X2}, [T||ID_B||PU_B])) \\ &= [T||ID_B||PU_B] \end{aligned} \quad (5)$$

PR_{X2} is X2's private key, ID_B is B's ID, PU_B is B's public key.

The above is only the core process in the simplest cross-domain authentication. When there is a complicated authentication path, the certificate issuance, update, and verification processes need to be repeated multiple times, which brings great challenges to authentication efficiency and certificate management overhead.

III. DIGITAL CERTIFICATE SCHEMA BASED ON BLOCKCHAIN

The traditional PKI coss-domain authentication model usually results in complex authentication path, multiple certificate signature and verification processes, certificate management difficulties and various security problems caused by certificate revocation. For addressing the problem, this paper proposes a digital certificate scheme based on blockchain. According to the X.509 standard, the certificate signing process is removed. Owing to the relevant characteristics of the blockchain, an efficient consensus algorithm is proposed for this scenario. The scheme realizes efficient coss-domain authentication of CA and secure revocation of digital certificate.

A. System Architecture

The root CA of each domain joins the blockchain network as a node. Under the given consensus algorithm, the efficient coss-domain authentication and the secure revocation of digital certificate in multi domain are realized through the blockchain transaction. The system architecture is shown in Figure 2. "Entity" is the entity in each domain, including users, resources, services, etc. AS is the authentication server of each domain, which is responsible for checking the digital certificate submitted by the user requesting communication.

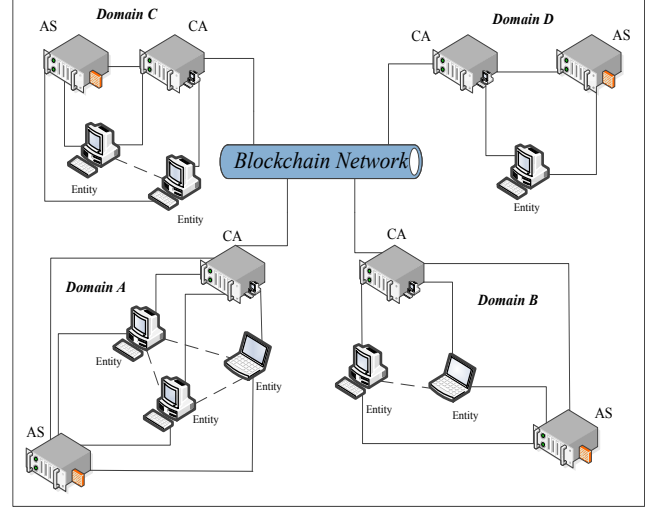


Figure 2. System Architecture.

B. Consensus Algorithm

The efficiency of coss-domain authentication and the timeliness of certificate revocation information sharing are very important in coss-domain communication. However, both well-known POW consensus algorithm and PBFT consensus algorithm can not meet the application in this scenario. The POW mechanism will cause a lot of waste of resources, and the transaction throughput under this mechanism is low, approximately 7TPS [13], which is far from being efficient. In the PBFT consensus process, $O(N^2)$ network messages are transmitted multiple times. In the case of view switching, $2n(n-1)+pn(n-1)$ times of communication are required, which causes a large network overhead and therefore the performance is not high [14].

The proposed consensus algorithm based on sharing hash of random numbers proposed has higher efficiency [15] in ensuring security, and the comparative analysis with PBFT consensus algorithm will be given in the next section. If a new transaction is initiated by a node in the blockchain network, the miner node needs to mine according to the consensus algorithm, and the successful miner is responsible for the generation of the new block, updating it into the blockchain ledger, and sharing it with all other nodes in the other blockchain. The specific process is as follows:

Consensus Algorithm

Parameters:

- N is the number of miners;
- m_i is miner, $1 \leq i \leq N$;
- λ means the mining difficulty, $0 < \lambda \leq 1$.

Process:

1. At the beginning of consensus period, each miner m_i generates a random number r_i locally.
2. The miner m_i calculates and broadcasts the hash value $H(r_i)$ of the generated random number r_i , H is the SHA256 hash function;
3. According to the difficulty coefficient λ , when a miner collects $[N\lambda]$ broadcasted hash values $\{H(r_0), H(r_1), \dots, H(r_{[N\lambda]})\}$, the miner broadcasts its own random number r_i ;
4. When a miner m_i collects $[N\lambda]$ broadcasted random number $\{r_0, r_1, \dots, r_{[N\lambda]}\}$, m_i calculates $num = \sum_{j=1}^{[N\lambda]} r_j \bmod [N\lambda]$;
5. Miner m_i is responsible for creating the next block containing the current transaction information.

The hash value of the certificate issued by the CA is included in the blockchain as a credential trusted by each domain. When cross-domain communication is carried out, the AS of each domain will query the CA hash value in the blockchain ledger. When a certificate needs to be revoked, the certificate hash value in the Certificate Revocation List will be included in the blockchain as the information shared among the domains in time. At the same time, the Certificate Revocation List will be stored in an open directory server accessible to each domain for the domain authentication server to query the status of the certificate through OCSP.

C. Cross-domain Authentication Protocol

Assume that user U_A in the domain A requests to access the user U_B in the domain B, and the flow chart of the authentication process is shown in Figure 3.

The details of each step are as follows:

Procedure of cross-domain authentication

1. $U_A \rightarrow AS_B$: the user U_A in the domain A requests to access the authentication server AS_B in the domain B.
2. $AS_B \xrightarrow{N} U_A$: The authentication server of domain B sends a random number of N to U_A as the certificate for the next information transfer.
3. $U_A \xrightarrow{\{Cert_A, E(PR_A, N)\}} AS_B$: U_A signs the received N with the private key and sends it to B with its own certificate.
4. $D(PU_A, E(PR_A, N))$: A checks whether the certificate $Cert_A$ is within the validity period. If not, the authentication fails. Otherwise, the signature information is verified based on the public key of U_A in the certificate, and N is correctly recovered. The hash value of the certificate is calculated for the next verification.
5. $AS_B \xrightarrow[\text{True/False}]{\text{Hash}(Cert_A)} Certificates$: AS_B Query whether the hash value of the certificate stored in the blockchain ledger contains the hash value of U_A 's certificate. If it exists, proceed to the next step, otherwise the authentication fails.
6. $AS_B \xrightarrow[\text{True/False}]{\text{Hash}(Cert_A)} CRLs$: Check whether the CRL contains the hash value of U_A 's certificate. If the certificate has not been revoked, proceed to the next step, otherwise the authentication fails.
7. $AS_B \xrightarrow{E(PR_{AS_B}, N_2)} U_A$: After the verification in the fifth and sixth steps, AS_B completes the authentication of U_A in domain A and allows it to access domain B. AS_B sends a random number N_2 encrypted by its private key to U_A as the certificate for U_A to communicate with other entities in B domain.
8. $U_A \xrightarrow{E(PR_{AS_B}, N_2)} U_B$: U_A sends U_B a message encrypted by the AS_B 's private key, and U_B can verify the message with the AS_B 's public key, it means that U_A has been successfully authenticated by AS_B and can establish communication with it.

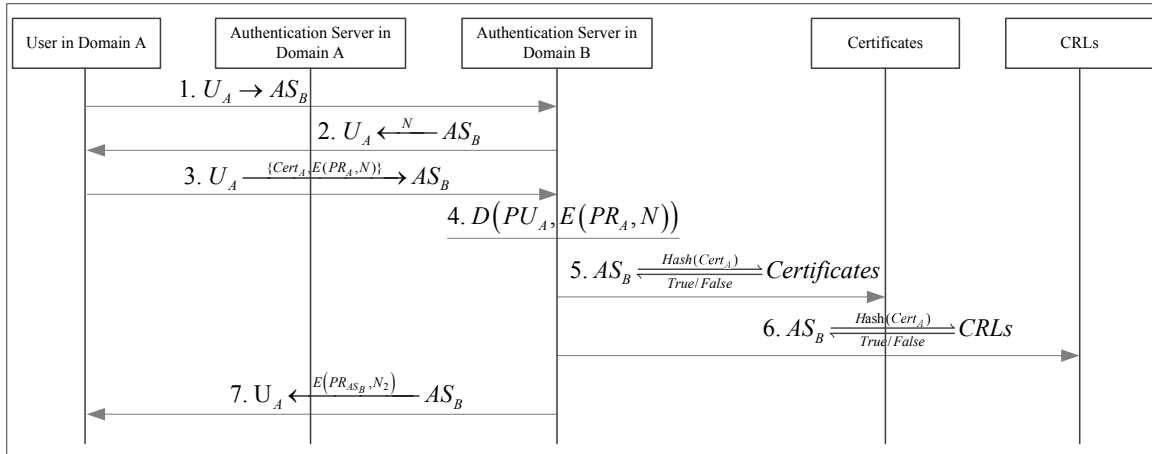


Figure 3. Cross-domain Authentication Protocol.

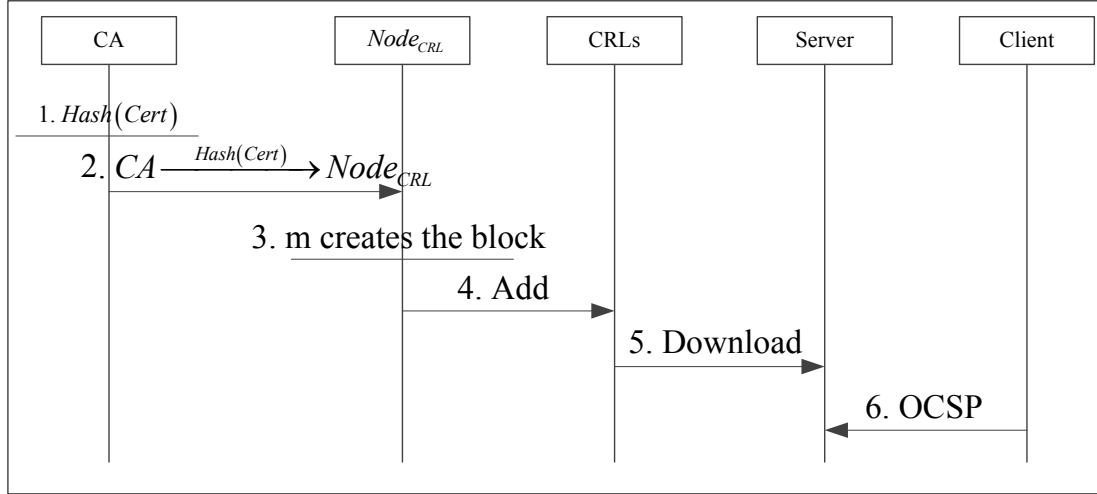


Figure 4. Certificate Revocation Protocol.

The cross-domain authentication process of the domain B to the domain A user U_A is given above. Similarly, the cross-domain authentication of the B domain user U_B by the domain A only requires the domain A's authentication server AS_A and U_B to repeat the above process.

D. Certificate Revocation

When the certificate is in the validity period, the CA will revoke the certificate in advance in the following cases [16], and record the certificate in the certificate revocation list:

- The user's private key is considered insecure;
- The certificate has been abandoned or not issued according to the rules of CA;
- The certificate is considered insecure.

Certificate revocation must ensure that certificate revocation list can be shared among multiple domains in time, and provide a fast query method at the same time. The certificate revocation flow chart is shown in Figure 4. The details of each step are as follows:

Process of certificate revocation

1. The CA detects that a certificate needs to be revoked in advance and calculates the hash value of the certificate.
2. The CA sends the hash value of the revocation certificate to the designated node $Node_{CRL}$ that receives the revocation information transaction.
3. According to the aforementioned consensus algorithm, miner m is responsible for the generation of transaction blocks, that is, the certificate of revocation is to be recorded in the blockchain ledger jointly maintained by each domain.
4. The revoked certificate record is added to the certificate revocation list maintained jointly for each domain to query.
5. The public directory server of each domain downloads the certificate revocation list to the local for the authentication server of each domain to query.
6. The client queries the status of the certificate through OCSP.

Based on blockchain transaction, the above certificate revocation protocol realizes the secure revocation of digital certificate through the common maintenance of CRL in multiple domains, efficient consensus algorithm and support of OCSP query.

IV. ANALYSIS OF SCHEMA

A. Security Analysis

• Non-repudiation

No matter cross-domain authentication or certificate revocation, the scheme takes the hash value of digital certificate as the trust certificate between domains, and compares the hash value of digital certificate provided by users with the hash value of certificate recorded in blockchain Ledger as the only judgment basis for authentication and revocation list query. The inherent properties of hash function ensure the non-repudiation of the scheme:

- 1) **Collision-resistance:** In the actual number range, for the same hash function $hash()$, two different digital certificates x_1 and x_2 have different hash values $hash(x_1)$ and $hash(x_2)$.
- 2) **One-way property:** Digital certificate x cannot be derived from its hash value $hash(x)$.

In addition, the blockchain is inherently undeniable. The hash values recorded in the blockchain ledger are all credible and undeniable. The above properties ensure that the scheme has non-repudiation.

• Anti DDoS attack

Unlike the centralized system, the blockchain-based scheme proposed in this paper naturally has features such as point-to-point and multiple redundancy. Even if one node fails, other nodes will not be affected, and there is no single point of failure.

The scheme in reference [12] uses decentralized consensus mechanism and introduces key sharing scheme to realize certificate revocation. In the scheme of reference

[12], it must be assumed that at least half of the miners can be trusted to ensure safety. The scheme in this paper has a higher security, because the final miner node responsible for generating a new block is jointly determined by the random numbers and hash values generated by all miners, so the selection of miners is completely random. Each node maintains a local transaction pool and completely randomly selects rules. Though the miner who is responsible for generating the block refuses to write a transaction into the block and only one miner is trusted, the trusted miner must be able to obtain the right to generate the block for a certain time. Therefore, the system security can be guaranteed finally.

• Anonymity

In this scheme, the block chain ledger and CRL server store not the digital certificate entity but the hash value of the digital certificate, so it does not contain the specific information of the entity. Because the hash function is unidirectional, the original certificate entity content can not be recovered according to the hash value of the digital certificate, which will not lead to privacy leaks. Therefore, this scheme has good anonymity and privacy[17].

B. Performance Analysis

• Analysis of communication cost

PBFT is the most commonly used consensus algorithm with high performance. In order to compare with the consensus algorithm proposed in this paper, suppose that there are n nodes in the blockchain network. In the case of PBFT considering view switching, the communication times of two consensus algorithms to complete a complete consensus are calculated respectively, as shown in Table I below.

TABLE I. COMPARISON OF COMMUNICATION COST OF CONSENSUS ALGORITHMS

Consensus Algorithm	Communication Cost
PBFT	$2n(n-1) + pn(n-1)$
Algorithm in this paper	$[2\lambda n(n-1), 2n(n-1)]$

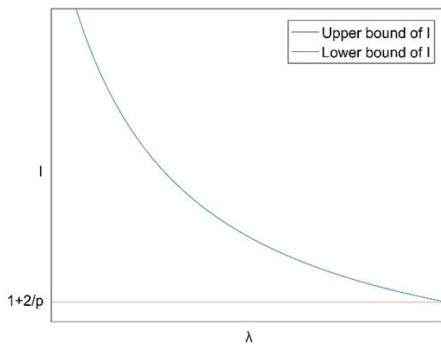


Figure 5. Comparison of communication cost of two consensus algorithms.

The algorithm in this paper completes a consensus with maximum communication cost of $2n(n-1)$ and minimum

communication cost of $2\lambda n(n-1)$. PBFT completes a consensus with communication cost of $2n(n-1) + pn(n-1)$, where p is the probability of view switching and λ is the difficulty coefficient of mining, $0 < p \leq 1$, $0 < \lambda \leq 1$. The communication cost ratio of the two schemes is I , $I \in [\frac{2+p}{2}, \frac{2+p}{2\lambda}]$, as shown in Figure 5.

Referring to the value range of p and λ , the value of I is always larger than 1, which indicates the scheme in this paper has lower communication cost.

• Analysis of computing cost

References [5] and [11] respectively proposed two typical improved cross-domain authentication schemes, the latter is also based on the using blockchain technology to implement cross-domain authentication scheme. In order to compare the efficiency of the scheme in this paper with that in [5] and [11], the computing cost in the same cross-domain scenario is calculated as shown in Table II.

TABLE II. COMPARISON OF COMPUTING COST OF CROSS-DOMAIN AUTHENTICATION

Scheme	Encryption times	Signature times	Hash calculation times
References [5]	0	12	4
References [11]	0	4	2
This paper	2	0	2

The scheme in [5] is based on elliptic curve threshold signature without trusted center, and the number of signatures will increase exponentially with the number of users, resulting in a significant decrease in efficiency. The cross-domain authentication protocol based on blockchain adopted in this paper does not need to sign the certificate, and the number of hash calculation is only half of the scheme in reference [5].

The cross-domain authentication scheme in reference [11] is based on the blockchain alliance chain technology, but the single authentication process needs to sign the digital certificate at least four times. In contrast, the proposed scheme does not need to sign the digital certificate and only needs to encrypt the random number twice, which increases the efficiency. Moreover, the blockchain consensus algorithm adopted in this scheme is more efficient than that in the alliance chain.

V. CONCLUSION AND FUTURE WORK

This paper addresses the problem of complex authentication paths during cross-domain authentication under the traditional PKI model, where multiple certificate signing and verification processes lead to low authentication efficiency and unsafe access. The blockchain can implement efficient cross-domain authentication and certificate security revocation schemes. Owing to the characteristics of the blockchain, the “signature” and “signature and algorithm identification” modules in the X.509 digital certificate are removed, and an efficient consensus algorithm is given to meet the efficiency and timeliness in this scenario. Finally, the specific cross-domain authentication and certificate

revocation protocols are proposed. While satisfying the efficiency, the scheme has good security, non-repudiation, anonymity, and resistance to DDos attacks. Compared with the PBFT algorithm, the consensus algorithm in this paper has lower communication cost. Through comparison and analysis with existing solutions, the proposed scheme has higher security and effectiveness.

The scheme proposed in this article is mainly focused on the scenario of cross-domain access on the web side. In the future, we should consider how to improve the solution to make it applicable in more network scenarios. In addition, the implementation of the scheme needs to build a blockchain network based on the proposed consensus algorithm. Future work will combine it with the existing blockchain network and use smart contracts for development to improve the convenience of development and maintenance.

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